Handling Missing Data in ETL – Assignment Answers

SECTION A – Theoretical Questions

Q1. What are the most common reasons for missing data in ETL pipelines?

Common causes include:

• Source-system data entry errors or optional fields.

• Data not collected for certain customers/transactions.

• Failed API, file or database extraction.

• Schema changes or source-column mismatches.

• Transformation/join errors that prevent values from matching.

• Data integration from multiple sources with different completeness levels.

• Network, timeout or ingestion failures.

• Privacy/business rules that intentionally suppress some values.

Q2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

Blind deletion can remove a large amount of useful information, introduce bias, and distort aggregates and KPIs. Missingness may not be random; for example, customers with missing income could belong to a particular segment. Deleting them can therefore change the customer population being analyzed. A better approach is to investigate the cause, assess the percentage and pattern of missingness, and choose an appropriate treatment such as imputation, flagging, source correction or controlled exclusion.

Q3. Difference between Listwise Deletion and Column Deletion

Listwise deletion removes an entire row/record when a specified value is missing. Example: removing customer records where Region is missing when Region is mandatory for a particular regional analysis.

Column deletion removes an entire column when that column has excessive missingness or provides little reliable analytical value. Example: dropping an optional source field when 95% of its values are missing and there is no reliable way to recover or impute it.

Listwise deletion affects records; column deletion affects a variable/field across the dataset.

Q4. Why is median imputation preferred over mean imputation for skewed data such as income?

Income is often right-skewed because a small number of customers may have very high incomes. The mean is strongly affected by these extreme values, whereas the median is much more resistant to outliers. Therefore, replacing missing income with the median generally gives a more representative central value for skewed income data.

Q5. What is forward fill and in what type of dataset is it most useful?

Forward fill replaces a missing value with the most recent previously available non-missing value. It is especially useful for ordered or time-series data where a value is expected to remain valid until a new value is recorded—for example, daily stock/status readings, sensor data, account status, or operational measurements. It should not be used blindly for independent customer attributes.

Q6. Why should flagging missing values be done before imputation in an ETL workflow?

A missing-value flag preserves information that the original value was missing. If the value is imputed first, that distinction can disappear. Creating a flag before imputation allows analysts and BI users to distinguish observed values from estimated values, measure the extent of missingness, and test whether missingness itself is associated with business outcomes.

Q7. How can missing income itself provide business insights?

A high level of missing income may indicate privacy concerns, low trust, incomplete onboarding, data-entry problems, or differences in how customer segments provide information. The pattern can reveal which regions, customer groups or acquisition channels have weaker data completeness. Businesses can use this insight to improve forms, collection processes and data-governance controls. Missingness should therefore be treated as a measurable data-quality/business signal, not merely as an inconvenience.

SECTION B – Practical Questions

Q8. Listwise Deletion – Remove rows where Region is missing

Affected row: Customer_ID 105 (Amit Verma, Pune), because Region is missing.

Records lost: 1

Dataset after deletion:

Customer_ID	Name	City	Monthly_Sales	Income	Region
101	Rahul Mehta	Mumbai	12000	65000	West
104	Neha Singh	Delhi	NaN	NaN	North
102	Anjali Rao	Bengaluru	NaN	NaN	South
107	Pooja Das	Kolkata	14000	NaN	East
103	Suresh Iyer	Chennai	15000	72000	South
106	Karan Shah	Ahmedabad	NaN	61000	West
108	Riya Kapoor	Jaipur	16000	69000	North

Q9. Imputation – Forward Fill for Monthly_Sales

Forward fill uses the previous available Monthly_Sales value. In the given row order, the missing values are filled as follows:

Customer_ID	Before	After Forward Fill	Explanation
101	12000	12000	Original value
104	NaN	12000	Uses previous value
102	NaN	12000	Uses previous value
105	18000	18000	Original value
107	14000	14000	Original value
103	15000	15000	Original value
106	NaN	15000	Uses previous value
108	16000	16000	Original value

Why suitable here: The question explicitly asks for forward fill, and the records have an ordered sequence. Forward fill is useful when Monthly_Sales is a time/order-dependent measure and the previous observation reasonably represents the missing period. However, for independent customer-level sales values, blindly carrying one customer's sales into another customer's row may be inappropriate; the business meaning of the row order must be confirmed first.

Q10. Flagging Missing Data – Income_Missing_Flag

Rule: Income_Missing_Flag = 0 when Income is present; 1 when Income is missing. In the supplied dataset, Neha Singh, Anjali Rao and Pooja Das have missing Income. Total missing-income customers = 3.

Customer_ID	Name	Income	Income_Missing_Flag
101	Rahul Mehta	65000	0
104	Neha Singh	NaN	1
102	Anjali Rao	NaN	1
105	Amit Verma	58000	0
107	Pooja Das	NaN	1
103	Suresh Iyer	72000	0
106	Karan Shah	61000	0
108	Riya Kapoor	69000	0

Final Summary

Missing data should be handled according to its cause, pattern and business meaning. In this dataset, listwise deletion removes one Region-missing record, forward fill produces the requested Monthly_Sales values based on row order, and an Income_Missing_Flag preserves the fact that three income values were originally missing.